

SDP

Note that SDP is a generalization of LP and there is a Poly time apx algorithm for SDP $(1 + \epsilon)\forall \epsilon > 0$ apx and the poly time is in size of input, $\ln \frac{1}{\epsilon}$

Define a matrix to be positive semidefinite: $Q \in R^{p \times p}$ is symmetric positive (semi)definite iff $\forall x \in R^p - \{0\}, x^T Q x \geq 0$ (semidefinite if $Q \geq 0$ and definite if $Q > 0$)

Thm $Q \geq 0$ iff symmetric all eigenvalues ≥ 0

($Q > 0$ iff symmetric all eigenvalues > 0)

Thm $Q \geq 0$ iff $\exists B : Q = B^T B$

Proof Sketch:

($\Leftarrow$) $Q = B^T B$ then $x^T Q x = x^T B^T B x = y^T y = |y|^2 \geq 0$ (note that $y = Bx$)

($\Rightarrow$) Incomplete Choleski factorization (or smth) that in poly time give $Q \rightarrow B$

$$B = [b_1, b_2, \dots, b_q]$$

$$Q = B^T B = [(b_1^T b_1, \dots, b_1^T b_q), (b_2^T b_1, \dots, b_2^T b_q), \dots, (b_q^T b_1, \dots, b_q^T b_q)]$$

Let $A, B \in R^{p \times q}$

$$A \cdot B = \sum_i^p \sum_j^q a_{ij} b_{ij}$$

SDP Standard Form

$$\min C \cdot X$$

$$\text{s.t. } A_i \cdot X = b_i (i = 1, 2, \dots, p)$$

$$X \geq 0$$

Note that $X \geq 0$ suggests that $\forall x x^T X \geq 0$, eigenvalues of X are ≥ 0 and $\exists B : B^T B = X$

WLOG X is symmetric

When trying to find B :

$$X = \begin{pmatrix} b_1^T b_1 & b_1^T b_2 & \dots & b_1^T b_h \\ b_2^T b_1 & b_2^T b_2 & \dots & b_2^T b_h \\ \dots & \dots & \dots & \dots \\ b_h^T b_1 & b_h^T b_2 & \dots & b_h^T b_h \end{pmatrix}$$

EX

$$C = [(1, -2), (-2, 4)]$$

$$A_1 = [(4, 3), (3, 1)], b_1 = 1$$

$$A_2 = [(1, 5), (5, 4)], b_2 = 2$$

$$X = [(x_1 1, x_1 2), (x_1 2, x_2 2)]$$

Which is equivalent to:

$$\min x_1 1 - 4x_1 2 + 4x_2 2$$

$$\text{s.t. } 4x_1 1 + 6x_1 2 + x_2 2 = 1$$

$$x_1 1 + 10x_1 2 + 4x_2 2 = 2$$

If we add a new constraint:

$A_3 = [(0, 1), (1, 0)], b_3 = 0$ which now forces $x_1 2$ to be 0 which now forces the SDP to become a linear program

So this A_3 turns the problem into a linear program because the eigenvalues are > 0 and they are the diagonal entries

Note to self figure out why SDP isnt by default a linear program if we dont add this A_3 constraint

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$$X = [(x_1 1, x_1 2), (x_1 2, x_2 2)]$$

$$\rightarrow \exists b_1, b_2 \in R^h : X = [(b_1^2, b_1^T b_2), (b_1^T b_2, b_2^2)]$$

If I wish to restrict: $b_1, b_2 \in S_h$ where S_h is a unit hypersphere